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```
In [1]: import os
import platform
print("=" * 100)
print("OS Name:", os.name)
print("System:", platform.system())
print("Release:", platform.release())
print("Machine:", platform.machine())
print("Processor:", platform.processor())
print("Platform:", platform.platform())

print("\nEnvironment Variables (sample):")
for key, value in list(os.environ.items())[:15]:
    print(f"{key} = {value}")
```

```
=====
OS Name: posix
System: Linux
Release: 4.18.0-348.12.2.el8_5.x86_64
Machine: x86_64
Processor: x86_64
Platform: Linux-4.18.0-348.12.2.el8_5.x86_64-x86_64-with-glibc2.28
```

```
Environment Variables (sample):
CONDA_SHLVL = 2
LD_LIBRARY_PATH = /usr/local/cuda/lib64:/usr/local/cuda/lib64:
CONDA_EXE = /work/tuan.tt19010226/miniconda/bin/conda
SSH_CONNECTION = 10.4.98.68 64873 10.20.8.11 22
CONDA_PREFIX = /work/tuan.tt19010226/conda/envs/RadarLM
XDG_SESSION_ID = 64511
USER = tuan.tt19010226
PWD = /home/tuan.tt19010226
HOME = /home/tuan.tt19010226
CONDA_PYTHON_EXE = /work/tuan.tt19010226/miniconda/bin/python
SSH_CLIENT = 10.4.98.68 64873 22
MESA_LOADER_DRIVER_OVERRIDE = llvmpipeexport
CONDA_PROMPT_MODIFIER = (RadarLM)
VSCODE_CLI_REQUIRE_TOKEN = b7ebbf12-935b-4329-b0c7-398a25100c8f
SHELL = /bin/bash
```

```
In [2]: platform.python_version()
```

```
Out[2]: '3.10.19'
```

Lab 3 - Changing Colorspaces

Changing Color-space

```
In [3]: import cv2
flags = [i for i in dir(cv2) if i.startswith('COLOR_')]
print(flags)
```

['COLOR_BAYER_BG2BGR', 'COLOR_BAYER_BG2BGRA', 'COLOR_BAYER_BG2BGR_EA', 'COLOR_BAYER_BG2BGR_VNG', 'COLOR_BAYER_BG2GRAY', 'COLOR_BAYER_BG2RGB', 'COLOR_BAYER_BG2RGBA', 'COLOR_BAYER_BG2RGB_EA', 'COLOR_BAYER_BG2RGB_VNG', 'COLOR_BAYER_BGR2BGR', 'COLOR_BAYER_BGR2BGRA', 'COLOR_BAYER_BGR2BGR_EA', 'COLOR_BAYER_BGR2BGR_VNG', 'COLOR_BAYER_BGR2GRAY', 'COLOR_BAYER_BGR2RGB', 'COLOR_BAYER_BGR2RGBA', 'COLOR_BAYER_BGR2RGB_EA', 'COLOR_BAYER_BGR2RGB_VNG', 'COLOR_BAYER_GB2BGR', 'COLOR_BAYER_GB2BGRA', 'COLOR_BAYER_GB2BGR_EA', 'COLOR_BAYER_GB2BGR_VNG', 'COLOR_BAYER_GB2GRAY', 'COLOR_BAYER_GB2RGB', 'COLOR_BAYER_GB2RGBA', 'COLOR_BAYER_GB2RGB_EA', 'COLOR_BAYER_GB2RGB_VNG', 'COLOR_BAYER_GBRG2BGR', 'COLOR_BAYER_GBRG2BGRA', 'COLOR_BAYER_GBRG2BGR_EA', 'COLOR_BAYER_GBRG2BGR_VNG', 'COLOR_BAYER_GBRG2GRAY', 'COLOR_BAYER_GBRG2RGB', 'COLOR_BAYER_GBRG2RGBA', 'COLOR_BAYER_GBRG2RGB_EA', 'COLOR_BAYER_GBRG2RGB_VNG', 'COLOR_BAYER_GR2BGR', 'COLOR_BAYER_GR2BGRA', 'COLOR_BAYER_GR2BGR_EA', 'COLOR_BAYER_GR2BGR_VNG', 'COLOR_BAYER_GR2GRAY', 'COLOR_BAYER_GR2RGB', 'COLOR_BAYER_GR2RGBA', 'COLOR_BAYER_GR2RGB_EA', 'COLOR_BAYER_GR2RGB_VNG', 'COLOR_BAYER_GRBG2BGR', 'COLOR_BAYER_GRBG2BGRA', 'COLOR_BAYER_GRBG2BGR_EA', 'COLOR_BAYER_GRBG2BGR_VNG', 'COLOR_BAYER_GRBG2GRAY', 'COLOR_BAYER_GRBG2RGB', 'COLOR_BAYER_GRBG2RGBA', 'COLOR_BAYER_GRBG2RGB_EA', 'COLOR_BAYER_GRBG2RGB_VNG', 'COLOR_BAYER_RG2BGR', 'COLOR_BAYER_RG2BGRA', 'COLOR_BAYER_RG2BGR_EA', 'COLOR_BAYER_RG2BGR_VNG', 'COLOR_BAYER_RG2GRAY', 'COLOR_BAYER_RG2RGB', 'COLOR_BAYER_RG2RGBA', 'COLOR_BAYER_RG2RGB_EA', 'COLOR_BAYER_RG2RGB_VNG', 'COLOR_BAYER_RGGB2BGR', 'COLOR_BAYER_RGGB2BGRA', 'COLOR_BAYER_RGGB2BGR_EA', 'COLOR_BAYER_RGGB2BGR_VNG', 'COLOR_BAYER_RGGB2GRAY', 'COLOR_BAYER_RGGB2RGB', 'COLOR_BAYER_RGGB2RGBA', 'COLOR_BAYER_RGGB2RGB_EA', 'COLOR_BAYER_RGGB2RGB_VNG', 'COLOR_BGR2BGR555', 'COLOR_BGR2BGR565', 'COLOR_BGR2BGRA', 'COLOR_BGR2GRAY', 'COLOR_BGR2HLS', 'COLOR_BGR2HLS_FULL', 'COLOR_BGR2HSV', 'COLOR_BGR2HSV_FULL', 'COLOR_BGR2LAB', 'COLOR_BGR2LUV', 'COLOR_BGR2Lab', 'COLOR_BGR2Luv', 'COLOR_BGR2RGB', 'COLOR_BGR2RGBA', 'COLOR_BGR2XYZ', 'COLOR_BGR2YCrCb', 'COLOR_BGR2YCrCb', 'COLOR_BGR2YUV', 'COLOR_BGR2YUV_I420', 'COLOR_BGR2YUV_IYUV', 'COLOR_BGR2YUV_UYUV', 'COLOR_BGR2YUV_UYVY', 'COLOR_BGR2YUV_YV12', 'COLOR_BGR2YUV_YVYU', 'COLOR_BGR552BGR', 'COLOR_BGR552BGRA', 'COLOR_BGR552GRAY', 'COLOR_BGR552RGB', 'COLOR_BGR552RGBA', 'COLOR_BGR5652BGR', 'COLOR_BGR5652BGRA', 'COLOR_BGR5652GRAY', 'COLOR_BGR5652RGB', 'COLOR_BGR5652RGBA', 'COLOR_BGRA2BGR', 'COLOR_BGRA2BGR555', 'COLOR_BGRA2BGR565', 'COLOR_BGRA2GRAY', 'COLOR_BGRA2RGB', 'COLOR_BGRA2RGBA', 'COLOR_BGRA2YUV_I420', 'COLOR_BGRA2YUV_IYUV', 'COLOR_BGRA2YUV_UYUV', 'COLOR_BGRA2YUV_UYVY', 'COLOR_BGRA2YUV_Y422', 'COLOR_BGRA2YUV_YUNV', 'COLOR_BGRA2YUV_YUY2', 'COLOR_BGRA2YUV_YUYV', 'COLOR_BGRA2YUV_YV12', 'COLOR_BGRA2YUV_YVYU', 'COLOR_BayerBG2BGR', 'COLOR_BayerBG2BGRA', 'COLOR_BayerBG2BGR_EA', 'COLOR_BayerBG2BGR_VNG', 'COLOR_BayerBG2GRAY', 'COLOR_BayerBG2RGB', 'COLOR_BayerBG2RGBA', 'COLOR_BayerBG2RGB_EA', 'COLOR_BayerBG2RGB_VNG', 'COLOR_BayerBGR2BGR', 'COLOR_BayerBGR2BGRA', 'COLOR_BayerBGR2BGR_EA', 'COLOR_BayerBGR2BGR_VNG', 'COLOR_BayerBGR2GRAY', 'COLOR_BayerBGR2RGB', 'COLOR_BayerBGR2RGBA', 'COLOR_BayerBGR2RGB_EA', 'COLOR_BayerBGR2RGB_VNG', 'COLOR_BayerBGRG2BGR', 'COLOR_BayerBGRG2BGRA', 'COLOR_BayerBGRG2BGR_EA', 'COLOR_BayerBGRG2BGR_VNG', 'COLOR_BayerBGRG2GRAY', 'COLOR_BayerBGRG2RGB', 'COLOR_BayerBGRG2RGBA', 'COLOR_BayerBGRG2RGB_EA', 'COLOR_BayerBGRG2RGB_VNG', 'COLOR_BayerGR2BGR', 'COLOR_BayerGR2BGRA', 'COLOR_BayerGR2BGR_EA', 'COLOR_BayerGR2BGR_VNG', 'COLOR_BayerGR2GRAY', 'COLOR_BayerGR2RGB', 'COLOR_BayerGR2RGBA', 'COLOR_BayerGR2RGB_EA', 'COLOR_BayerGR2RGB_VNG', 'COLOR_BayerGRBG2BGR', 'COLOR_BayerGRBG2BGRA', 'COLOR_BayerGRBG2BGR_EA', 'COLOR_BayerGRBG2BGR_VNG', 'COLOR_BayerGRBG2GRAY', 'COLOR_BayerGRBG2RGB', 'COLOR_BayerGRBG2RGBA', 'COLOR_BayerGRBG2RGB_EA', 'COLOR_BayerGRBG2RGB_VNG', 'COLOR_BayerGRB2BGR', 'COLOR_BayerGRB2BGRA', 'COLOR_BayerGRB2BGR_EA', 'COLOR_BayerGRB2BGR_VNG', 'COLOR_BayerRG2BGR', 'COLOR_BayerRG2BGRA', 'COLOR_BayerRG2BGR_EA', 'COLOR_BayerRG2BGR_VNG', 'COLOR_BayerRG2GRAY', 'COLOR_BayerRG2RGB', 'COLOR_BayerRG2RGBA', 'COLOR_BayerRG2RGB_EA', 'COLOR_BayerRG2RGB_VNG', 'COLOR_BayerRGG2BGR', 'COLOR_BayerRGG2BGRA', 'COLOR_BayerRGG2BGR_EA', 'COLOR_BayerRGG2BGR_VNG', 'COLOR_BayerRGG2GRAY', 'COLOR_BayerRGG2RGB', 'COLOR_BayerRGG2RGBA', 'COLOR_BayerRGG2RGB_EA', 'COLOR_BayerRGG2RGB_VNG', 'COLOR_BayerRGGB2BGR', 'COLOR_BayerRGGB2BGRA', 'COLOR_BayerRGGB2BGR_EA', 'COLOR_BayerRGGB2BGR_VNG', 'COLOR_BayerRGGB2GRAY', 'COLOR_BayerRGGB2RGB', 'COLOR_BayerRGGB2RGBA', 'COLOR_BayerRGGB2RGB_EA', 'COLOR_BayerRGGB2RGB_VNG', 'COLOR_COLORCVT_MAX', 'COLOR_GRAY2BGR', 'COLOR_GRAY2BGR555', 'COLOR_GRAY2BGR565', 'COLOR_GRAY2BGRA', 'COLOR_GRAY2RGB', 'COLOR_GRAY2RGBA', 'COLOR_HLS2BGR', 'COLOR_HLS2BGR_FULL', 'COLOR_HLS2RGB', 'COLOR_HLS2RGB_FULL', 'COLOR_HSV2BGR', 'COLOR_HSV2BGR_FULL', 'COLOR_HSV2RGB', 'COLOR_HSV2RGB_FULL', 'COLOR_LAB2BGR', 'COLOR_LAB2LBGR', 'COLOR_LAB2LRGB', 'COLOR_LAB2RGB', 'COLOR_LBGR2LAB', 'COLOR_LBGR2LUV', 'COLOR_LBGR2Lab', 'COLOR_LBGR2Luv', 'COLOR_LRGB2LAB', 'COLOR_LRGB2LUV', 'COLOR_LRGB2Lab', 'COLOR_LRGB2Luv', 'COLOR_LUV2BGR', 'COLOR_LUV2LBGR', 'COLOR_LUV2LRGB', 'COLOR_LUV2RGB', 'COLOR_Lab2BGR', 'COLOR_Lab2LBGR', 'COLOR_Lab2LRGB', 'COLOR_Lab2RGB', 'COLOR_Luv2BGR', 'COLOR_Luv2LBGR', 'COLOR_Luv2LRGB', 'COLOR_Luv2RGB', 'COLOR_M_RGBA2RGBA', 'COLOR_RGB2BGR', 'COLOR_RGB2BGR555', 'COLOR_RGB2BGR565', 'COLOR_RGB2BGRA', 'COLOR_RGB2GRAY', 'COLOR_RGB2HLS', 'COLOR_RGB2HLS_FULL', 'COLOR_RGB2HSV', 'COLOR_RGB2HSV_FULL', 'COLOR_RGB2LAB', 'COLOR_RGB2LUV', 'COLOR_RGB2Lab', 'COLOR_RGB2Luv', 'COLOR_RGB2RGBA', 'COLOR_RGB2XYZ', 'COLOR_RGB2YCrCb', 'COLOR_RGB2YCrCb', 'COLOR_RGB2YUV', 'COLOR_RGB2YUV_I420', 'COLOR_RGB2YUV_IYUV', 'COLOR_RGB2YUV_UYUV', 'COLOR_RGB2YUV_UYVY', 'COLOR_RGB2YUV_Y422', 'COLOR_RGB2YUV_YUNV', 'COLOR_RGB2YUV_YUY2', 'COLOR_RGB2YUV_YUYV', 'COLOR_RGB2YUV_YV12', 'COLOR_RGB2YUV_YVYU', 'COLOR_RGBA2BGR', 'COLOR_RGBA2BGR555', 'COLOR_RGBA2BGR565', 'COLOR_RGBA2BGRA', 'COLOR_RGBA2GRAY', 'COLOR_RGBA2M_RGBA', 'COLOR_RGBA2RGB', 'COLOR_RGBA2YUV_I420', 'COLOR_RGBA2YUV_IYUV', 'COLOR_RGBA2YUV_UYUV', 'COLOR_RGBA2YUV_UYVY', 'COLOR_RGBA2YUV_Y422', 'COLOR_RGBA2YUV_YUNV', 'COLOR_RGBA2YUV_YUY2', 'COLOR_RGBA2YUV_YUYV', 'COLOR_RGBA2YUV_YV12', 'COLOR_RGBA2YUV_YVYU', 'COLOR_RGBA2mRGBA', 'COLOR_XYZ2BGR', 'COLOR_XYZ2RGB', 'COLOR_YCrCb2BGR', 'COLOR_YCrCb2BGR_EA', 'COLOR_YCrCb2BGR_VNG', 'COLOR_YUV2BGR', 'COLOR_YUV2BGRA_I420', 'COLOR_YUV2BGRA_IYUV', 'COLOR_YUV2BGRA_NV12', 'COLOR_YUV2BGRA_NV21', 'COLOR_YUV2BGRA_UYUV', 'COLOR_YUV2BGRA_UYVY', 'COLOR_YUV2BGRA_Y422', 'COLOR_YUV2BGRA_YUNV', 'COLOR_YUV2BGRA_YUY2', 'COLOR_YUV2BGRA_YUYV', 'COLOR_YUV2BGRA_YV12', 'COLOR_YUV2BGRA_YVYU', 'COLOR_YUV2BGR_I420', 'COLOR_YUV2BGR_IYUV', 'COLOR_YUV2BGR_NV12', 'COLOR_YUV2BGR_NV21', 'COLOR_YUV2BGR_UYUV', 'COLOR_YUV2BGR_UYVY', 'COLOR_YUV2BGR_Y422', 'COLOR_YUV2BGR_YUNV', 'COLOR_YUV2BGR_YUY2', 'COLOR_YUV2BGR_YUYV', 'COLOR_YUV2GRAY_420', 'COLOR_YUV2GRAY_I420', 'COLOR_YUV2GRAY_IYUV', 'COLOR_YUV2GRAY_NV12', 'COLOR_YUV2GRAY_V2GRAY_NV21', 'COLOR_YUV2GRAY_UYUV', 'COLOR_YUV2GRAY_UYVY', 'COLOR_YUV2GRAY_Y422', 'COLOR_YUV2GRAY_YUNV', 'COLOR_YUV2GRAY_YUY2', 'COLOR_YUV2GRAY_YUYV', 'COLOR_YUV2GRAY_YV12', 'COLOR_YUV2GRAY_YVYU', 'COLOR_YUV2RGB', 'COLOR_YUV2RGBA_I420', 'COLOR_YUV2RGBA_IYUV', 'COLOR_YUV2RGBA_NV12', 'COLOR_YUV2RGBA_NV21', 'COLOR_YUV2RGBA_UYUV', 'COLOR_YUV2RGBA_UYVY', 'COLOR_YUV2RGBA_Y422', 'COLOR_YUV2RGBA_YUNV', 'COLOR_YUV2RGBA_YUY2', 'COLOR_YUV2RGBA_YUYV', 'COLOR_YUV2RGB_YV12', 'COLOR_YUV2RGB_YVYU', 'COLOR_YUV420P2BGR', 'COLOR_YUV420P2BGRA', 'COLOR_YUV420P2GRAY', 'COLOR_YUV420P2RGB', 'COLOR_YUV420P2RGBA', 'COLOR_YUV420SP2BGR', 'COLOR_YUV420SP2BGRA', 'COLOR_YUV420SP2GRAY', 'COLOR_YUV420SP2RGB', 'COLOR_YUV420SP2RGBA', 'COLOR_YUV420p2BGR', 'COLOR_YUV420p2BGRA', 'COLOR_YUV420p2GRAY', 'COLOR_YUV420p2RGB', 'COLOR_YUV420p2RGBA', 'COLOR_YUV420sp2BGR', 'COLOR_YUV420sp2BGRA', 'COLOR_YUV420sp2GRAY', 'COLOR_YUV420sp2RGB', 'COLOR_YUV420sp2RGBA', 'COLOR_mRGBA2RGBA']

Object Tracking

```
In [ ]: import cv2 as cv
import numpy as np

cap = cv.VideoCapture(0)
```

```

while(1):

    # Take each frame
    _, frame = cap.read()

    # Convert BGR to HSV
    hsv = cv.cvtColor(frame, cv.COLOR_BGR2HSV)

    # define range of blue color in HSV
    lower_blue = np.array([110,50,50])
    upper_blue = np.array([130,255,255])

    # Threshold the HSV image to get only blue colors
    mask = cv.inRange(hsv, lower_blue, upper_blue)

    # Bitwise-AND mask and original image
    res = cv.bitwise_and(frame,frame, mask= mask)

    cv.imshow('frame',frame)
    cv.imshow('mask',mask)
    cv.imshow('res',res)
    k = cv.waitKey(5) & 0xFF
    if k == 27:
        break

cv.destroyAllWindows()

```

How to find HSV values to track?

```

In [5]: import numpy as np

green = np.uint8([[0,255,0 ]])
hsv_green = cv2.cvtColor(green,cv2.COLOR_BGR2HSV)
print(hsv_green)

[[[ 60 255 255]]]

```

```

In [6]: import urllib.request
import matplotlib.pyplot as plt

url = "https://chainhuahuyhat.com/wp-content/uploads/2021/01/cac-loai-nap-chai-nhua.jpg"

resp = urllib.request.urlopen(url)
img = np.asarray(bytearray(resp.read()), dtype=np.uint8)
img = cv2.imdecode(img, cv2.IMREAD_COLOR)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
plt.imshow(img_rgb)
plt.axis(False)
plt.show()

```



```

In [7]: hsv = cv2.cvtColor(img_rgb, cv2.COLOR_RGB2HSV)

# ----- BLUE -----
lower_blue = np.array([100, 150, 50])
upper_blue = np.array([130, 255, 255])

mask_blue = cv2.inRange(hsv, lower_blue, upper_blue)

```

```

# ----- GREEN -----
lower_green = np.array([40, 50, 50])
upper_green = np.array([80, 255, 255])

mask_green = cv2.inRange(hsv, lower_green, upper_green)

# ----- RED -----
lower_red1 = np.array([0, 120, 70])
upper_red1 = np.array([10, 255, 255])

lower_red2 = np.array([170, 120, 70])
upper_red2 = np.array([180, 255, 255])

mask_red1 = cv2.inRange(hsv, lower_red1, upper_red1)
mask_red2 = cv2.inRange(hsv, lower_red2, upper_red2)

mask_red = mask_red1 + mask_red2
combined_mask = mask_blue + mask_green + mask_red
result = cv2.bitwise_and(img_rgb, img_rgb, mask=combined_mask)
plt.figure(figsize=(12, 8))

plt.subplot(2, 3, 1)
plt.imshow(img_rgb)
plt.title("Original")
plt.axis("off")

plt.subplot(2, 3, 2)
plt.imshow(mask_blue, cmap="gray")
plt.title("Blue Mask")
plt.axis("off")

plt.subplot(2, 3, 3)
plt.imshow(mask_green, cmap="gray")
plt.title("Green Mask")
plt.axis("off")

plt.subplot(2, 3, 4)
plt.imshow(mask_red, cmap="gray")
plt.title("Red Mask")
plt.axis("off")

plt.subplot(2, 3, 5)
plt.imshow(combined_mask, cmap="gray")
plt.title("Combined Mask")
plt.axis("off")

plt.subplot(2, 3, 6)
plt.imshow(result)
plt.title("Result")
plt.axis("off")

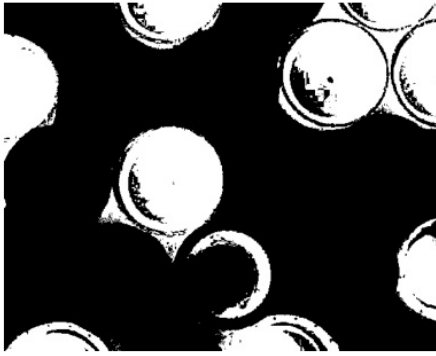
plt.tight_layout()
plt.show()

```

Original



Blue Mask



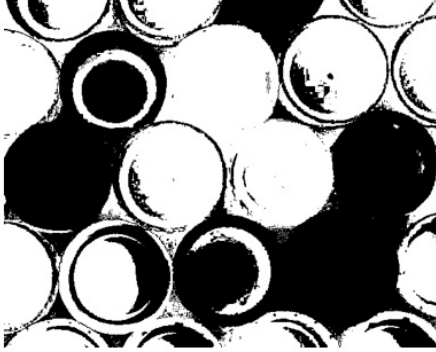
Green Mask



Red Mask



Combined Mask



Result

